

PLASMA METABOLOMIC PROFILING TO IDENTIFY METABOLITES ASSOCIATED WITH HYPERINSULINAEMIA

Josephine Vovos¹, Shaun Mason¹, Chris Shaw¹ and Clinton Bruce¹

¹ Institute for Physical Activity and Nutrition, Deakin University

Email: s218257401@deakin.edu.au

INTRODUCTION

Insulin is responsible for maintaining blood glucose concentrations within a narrow range. Hyperinsulinaemia, characterised by chronically elevated circulating insulin, is commonly associated with obesity, insulin resistance (IR) and cardiometabolic diseases. However, emerging evidence suggests that hyperinsulinaemia may precede IR and weight gain. It can also occur despite normal glucose tolerance (NGT), suggesting that factors beyond glucose may contribute to an exaggerated insulin response. Understanding these factors may provide insight into the development of hyperinsulinaemia independent of obesity and dysglycaemia. Recent work from our research group identified a novel phenotype termed isolated postprandial hyperinsulinaemia (IPH) [1]. This group was characterised by normal fasting insulin concentrations but an exaggerated postprandial insulin response. The mechanisms underlying the differences between fasting and postprandial hyperinsulinaemia remain poorly understood. Therefore, this study utilised a population of young (18–25 years), non-obese (BMI <30 kg/m²) adults with NGT to determine whether IPH is associated with a distinct metabolic signature, and to identify metabolites associated with hyperinsulinaemia beyond glucose.

METHODS

This study utilised existing data and plasma samples collected as part of a previous PhD project [2]. The original study (n=147) involved rigorous participant screening including anthropometric measurements, dietary and physical activity assessment and metabolic and cardiovascular health evaluations. Participants underwent a standard 2-hour oral glucose tolerance test (OGTT), with plasma insulin measured at multiple time points. Based on their insulin responses, participants

were categorised into hyperinsulinaemic phenotypes, with the OGTT repeated to confirm phenotype classification.

Here, plasma samples collected during the second OGTT were analysed using untargeted metabolomic profiling using high performance liquid chromatography coupled to high-resolution Orbitrap mass spectrometry (LC-MS). Metabolomic profiles were analysed using linear mixed models to assess the effects of time, group, and the group × time interaction.

RESULTS AND DISCUSSION

A total of 47 participants (57% female, 26 (4) years) were included. Based on fasting and postprandial insulin responses, 36% were classified as normoinsulinaemic (NI), 30% as isolated postprandial hyperinsulinaemic (IPH), and 34% as overt hyperinsulinaemic (HI), characterised by elevated fasting and postprandial insulin concentrations (Table 1).

LC-MS analysis has been completed, identifying more than 300 unique plasma metabolites. Data analysis is currently underway to identify potential differences in metabolomic profiles between the hyperinsulinaemic phenotypes and determine metabolites associated with hyperinsulinaemia.

CONCLUSIONS

While this study is currently incomplete, its findings may provide novel insight into the metabolomic profiles associated with hyperinsulinaemia in young, healthy NGT individuals.

REFERENCES

- [1] Hamley S. *Deakin University* [PhD Thesis]: 2020.
- [2] Kuriel V. *Deakin University* [PhD Thesis]: 2025.

Table 1: Fasting and postprandial insulin responses for NI, IPH and HI groups

	NI	IPH	HI	p-value
Fasting insulin (pmol/l)	27.8±9.5	34.9±10.5	75.7±42.6	$p < 0.05$
Insulin iAUC (pmol/l x120 min)	27964±9992	64630±23143	74151±28293	$p < 0.05$

Data are (actual) mean ± SD reported.